
NTPServer

A GPS-Disciplined Stratum-1 Network Time Server Operator Manual

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This manual covers the setup, configuration, and day-to-day operation of the NTPServer — a self-contained network time server that derives time from a GPS receiver (or an upstream Internet time server), disciplines a temperature-compensated DS3231 real-time clock to that source, and serves it to NTP clients on your local network. It includes a Theory of Operation chapter explaining how the server produces and distributes accurate time.

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Quick Start

The NTPServer is a stratum-1 network time server. It keeps highly accurate time from a GPS receiver (or, as a fallback, from an Internet time server) and hands that time out to every computer, clock, and device on your network using the standard NTP protocol. A full explanation of every option is in the numbered sections that follow.

1. Position the server and power it

The GPS antenna is built into the receiver, so there is nothing to attach — just place the server itself where it can see the sky, such as a window sill or, ideally, outdoors. Connect a 5 V USB-C power supply. On a brand-new unit, AP setup mode starts automatically (see step 2).

2. Connect to the server's WiFi

From a phone or computer, open WiFi settings and look for the network named after the server — NTPServer by default. Connect to it; no password is required for the setup network.

3. Configure in a browser

Open a web browser and go to 192.168.4.1. The setup page loads. At minimum:

- Select your home WiFi network from the SSID dropdown and enter its password.
- Leave the Discipline Mode at NMEA GPS unless you intend to run from an Internet time server instead of GPS.

Tap Save, then Start. The server saves the settings and reboots onto your network.

4. Wait for a GPS fix

After reboot the server joins your WiFi and begins serving as soon as a time source is locked. A first GPS fix from a cold start can take from under a minute to several minutes depending on how much open sky the unit can see. The STATUS LED shows progress (Section 8), and the status web page shows the live state.

5. Point your devices at it

On each device, set the NTP server address to this unit's IP address (shown on the status page) or its mDNS name — ntpserver.local by default, or <name>.local if you

change the Access Point Name. The server answers standard NTP requests on UDP port 123 — no client software is required beyond the NTP client already built into every operating system.

6. Daily use

To do this	Do this
See live status	Browse to the server's IP address (or http://ntpserver.local/) from any device on the same WiFi
Show the IP address on the LEDs	Double-click SW1 — the IP blinks out octet by octet
Install new software over WiFi	Hold SW1 for 10 seconds
Re-enter setup (AP) mode	Hold SW1 for 5 seconds, or use Switch to AP Mode on the status page (System sub-page)
Factory reset	Hold SW1 while applying power; the unit wipes its settings and reboots into AP mode

1. Overview

The NTPServer is a self-contained network time server built on an ESP32-S3. It derives time from one of two sources — a u-blox GPS receiver or an upstream Internet NTP server — disciplines a temperature-compensated DS3231 real-time clock to that source, and serves the result to NTP clients on your local network. A web interface handles all configuration; there is no app to install and no cloud account.

Why run your own time server?

- Accuracy. A GPS-disciplined server delivers time within roughly a millisecond of true UTC on your LAN — far tighter than typical Internet sync, and with no dependence on your Internet connection once GPS is locked.
- Independence. Your devices sync to a source on your own network. They stay in agreement with each other even if the Internet is down.
- Privacy and control. NTP requests stay on your LAN rather than going out to public time servers.

What's in the box

- The NTPServer unit (ESP32-S3 controller, DS3231 RTC with battery backup, indicator and under-light LEDs).
- A u-blox GPS receiver with a built-in antenna and a 1PPS (one-pulse-per-second) output.
- A USB-C port for power — use any standard 5 V USB-C supply (power supply not included).

Key features

- Stratum-1 GPS discipline using NMEA time plus the 1PPS edge for sub-second accuracy.
- Automatic NTP fallback if GPS is unavailable, with a configurable holdover window.
- Battery-backed DS3231 RTC carries time through brief power and signal interruptions.
- Web-based setup and a live status page; mDNS discovery as ntpserver.local by default (follows the Access Point Name).
- Over-the-air software updates.
- Two status indicators plus two color-configurable under-lights.

2. Theory of Operation

This section explains how the NTPServer produces and distributes accurate time. It is background reading — none of it is required to use the server, but it explains what the status page is telling you and why the design behaves as it does.

The problem NTP solves

Every computer has a clock, but those clocks drift — a few seconds a day is typical. The Network Time Protocol (NTP) lets a device ask a time server "what time is it?", measure how long the question-and-answer round trip took, and correct its own clock accordingly. The key trick is that NTP timestamps the request on the way out and the reply on the way back, so the client can subtract the network delay and recover the true time the server held at a known instant. The quality of everyone's clock therefore depends on the quality of the server's clock — which is what this device is built to maximize.

Stratum: the time hierarchy

NTP arranges time sources in numbered tiers called strata. Stratum 0 is a reference clock itself — an atomic clock or, here, the GPS system, whose satellites carry atomic clocks. A server directly attached to a stratum-0 reference is stratum 1. A server that syncs from a stratum-1 server is stratum 2, and so on; each hop adds a little uncertainty. When this unit is GPS-disciplined it advertises stratum 1. When it falls back to an Internet NTP server (typically stratum 2) it advertises stratum 3.

The time source: a disciplined RTC

At the heart of the server is a DS3231 — a temperature-compensated real-time clock chip accurate to roughly two parts per million (about a tenth of a second per day on its own). The DS3231 is not trusted to keep perfect time by itself; instead it is continuously corrected — "disciplined" — by a more accurate source. Between corrections, the server does not read the chip's ticking registers for the served time. It keeps the corrected moment in memory and extrapolates forward using the processor's own millisecond timer, which yields a smooth, phase-accurate clock that can report not just the second but the precise fraction of a second.

Discipline source 1: GPS

In the default mode the discipline source is the GPS receiver. GPS is two signals working together. First, the receiver sends NMEA sentences over a serial line —

ordinary text messages that include the current date and time to the whole second. Second, and crucially, it emits a 1PPS pulse: a sharp electrical edge that occurs exactly on each UTC second boundary, accurate to tens of nanoseconds. The NMEA text says which second it is; the 1PPS edge says exactly when that second begins. The firmware combines the two — taking the whole second from the text and the precise instant from the pulse — to anchor the served clock to true UTC within well under a millisecond. This is what makes the server stratum 1.

Discipline source 2: upstream NTP

If you are running without GPS, or GPS never acquires a fix, the server can instead discipline itself from an Internet NTP server (for example `pool.ntp.org` or `time.google.com`). This is less accurate — Internet round trips introduce some tens of milliseconds of uncertainty — but it needs no view of the sky and still produces a perfectly usable LAN time server. In this mode the unit advertises stratum 3.

Holdover: riding through a GPS dropout

GPS reception is not perfectly continuous — a passing obstruction or a poor view of the sky can briefly interrupt the fix. Because the DS3231 drifts only about seven milliseconds per hour, the server does not panic the instant GPS disappears. It enters holdover: it keeps serving GPS-derived time from the DS3231, still as stratum 1, for a configurable window (one hour to one day, default four hours). Only if GPS stays gone past that window does the server hand discipline to the NTP fallback. The logic deliberately prefers a few hours of holdover over an immediate switch, because the drifting-but-recent GPS time is actually tighter than a fresh Internet NTP sample would be.

NTP as a witness

When both GPS and an upstream NTP server are available, the NTP fallback does not sit idle. While GPS holds discipline, the server still polls the upstream NTP server on a schedule — but only to compare, not to correct. The difference between the two independent sources is shown on the status page as the GPS–NTP figure. In normal operation it is a few milliseconds, which is reassuring evidence that two unrelated time sources agree. If they ever disagree by more than a second and keep disagreeing, the figure turns red — a genuine alarm that something (a GPS fault, poor reception, a misbehaving upstream server) deserves attention.

Guarding against bad data

A time server that blindly accepted every sample could be thrown wildly off by a single garbled GPS sentence or a malformed packet. Two safeguards prevent this. First, GPS sentences are checksum-verified before use, and NTP responses are validated against the protocol. Second, a sanity filter rejects any correction that would jump the clock by more than a minute — unless the same large offset persists across several samples, in

which case it is eventually accepted as a real (if dramatic) correction. The result is a clock that tracks small, legitimate adjustments instantly but refuses to be yanked around by a one-off glitch.

Serving clients

The server listens for NTP requests on UDP port 123. When a request arrives it records the exact moment of arrival, fills in the reply with its reference identifier and stratum, stamps the precise transmit time, and sends it back. One safeguard is fundamental: the server refuses to answer at all until it has actually locked to a real time source. Before the first GPS or NTP sync it stays silent rather than handing out unverified, free-running RTC time that a client would trust. The moment a source locks, serving begins automatically.

Putting it together

On a typical boot the sequence is: the DS3231 provides a rough starting time from its backup battery (but the server does not serve yet); WiFi connects and, if configured, an NTP sample disciplines the clock within seconds; GPS then acquires a fix and takes over as the stratum-1 source, with NTP demoted to witness. From then on the served clock is anchored to the GPS 1PPS edge every second, holds over through brief GPS dropouts, and is continuously cross-checked against an independent Internet source — all while answering client requests on the LAN.

3. Hardware

The server is a single board built around an ESP32-S3 module. The major components and their roles:

Component	Role
ESP32-S3 controller	Runs the firmware: WiFi, NTP server, GPS parsing, web interface, LED control
DS3231 RTC	Temperature-compensated real-time clock with coin-cell backup; the disciplined time source
u-blox GPS receiver	Supplies NMEA time messages and the 1PPS one-second pulse
4 × APA102 LEDs	Two status indicators (top) and two color-configurable under-lights (bottom)
SW1 button	IP-address readout (double-click), AP mode (5-second hold), and over-the-air update (10-second hold)
USB-C	5 V power input

A coin cell on the DS3231 preserves the time across power loss, so after a reboot the server starts from a close approximation and is back to full accuracy as soon as a source re-locks.

4. First-Time Setup (AP Mode)

A brand-new server, or one that has been factory reset, has no saved WiFi credentials and so starts in Access Point (AP) setup mode automatically. In this mode the two status LEDs alternate steadily.

Connecting

From a phone or computer, open WiFi settings and join the network named NTPServer (or whatever Access Point Name has been configured). No password is required. Then open a browser to 192.168.4.1.

Network settings

- SSID — choose your 2.4 GHz home network from the dropdown. The ESP32 does not support 5 GHz networks.
- SSID Override — optionally type a network name by hand (for a hidden network, or one not in range during setup).
- Network Password — the password for the selected network.
- Access Point Name — the name this server advertises, used both for the AP network and as its mDNS name (<name>.local).

Static IP (optional)

By default the server takes an address from your router by DHCP. Because clients are pointed at this server by address, a stable address is helpful; the most reliable approach is a DHCP reservation in your router. Alternatively, enable Use Static IP and fill in the IP address, gateway, subnet mask, and DNS server — choose an address outside your router's DHCP pool.

Finishing

Tap Save to store settings without rebooting, or Start to commit and reboot onto your network. After Start, reconnect your phone or computer to your normal WiFi; the server will appear at its DHCP or static address and as ntpserver.local.

5. The Status Page

Browsing to the server's address shows a live status page that refreshes once per second without reloading. It is the primary window into what the server is doing.

Field	Meaning
UTC	The current served time in Coordinated Universal Time
Source	The active discipline source: GPS, NTP, or — (none yet)
Status	LOCKED (serving), HOLDOVER (GPS lost but riding the RTC), or NO SYNC
Last sync	How long ago the active source last corrected the clock
DS3231 temp	The RTC chip temperature (Celsius or Fahrenheit per Settings)
WiFi	Connected network and signal strength, or AP mode
NTP	The server's own IP address on your network
mDNS	The <name>.local address for discovery
Served	Lifetime count of NTP requests answered, and the last client address
Uptime	Time since the server last achieved lock
Heap	Free memory and fragmentation (diagnostic)
Last reset	Why the server last restarted: power-on, software, panic, etc. (diagnostic)
GPS-NTP	In GPS mode: the difference between GPS and the upstream NTP server, in milliseconds, while both are live (a dash otherwise). Turns red on a persistent disagreement over one second.

6. Settings Reference

These options appear on the setup form and, where they apply at runtime, on the live settings page. Changes on the live page take effect immediately; Save persists them.

Time source

Setting	What it does
Discipline Mode	NMEA GPS (default; GPS primary, NTP fallback) or Upstream NTP (no GPS; NTP is the only source)
NTP Server	Primary upstream time server, used as the fallback in GPS mode or the source in NTP mode
Alternate NTP Server	A second server the firmware rotates to if the primary is unreachable
NTP Polling Interval	How often to query the upstream NTP server (5 minutes to 4 hours)
NTP Fallback	How long GPS keeps discipline through a dropout before NTP takes over (1 hour to 1 day; default 4 hours)
GPS Baud	Serial speed of the GPS module (this receiver ships at 9600)

Display and LEDs

Setting	What it does
Temperature Units	Celsius or Fahrenheit for the DS3231 temperature on the status page
Indicator LEDs	Brightness of the two top status LEDs: Disabled, Low, Medium, or High
Under-light Color	The color of the two bottom LEDs. The color also sets their brightness — a darker shade dims them, black turns them off

System

The System sub-page (available once the server is on your network) hosts the Firmware URL, the Update Firmware button, and Switch to AP Mode.

7. Indicator LEDs and Under-light

The server has four LEDs in one chain: two status indicators on top and two under-lights on the bottom.

Status LEDs (top)

The two top LEDs report what the server is doing. The STATUS LED's color encodes the active time source, and its blink pattern encodes the lock state; the ACTIVITY LED flashes as requests are served.

State	STATUS LED	ACTIVITY LED
GPS locked	● Blue — one short heartbeat pulse per second	White flash on each NTP request served
NTP locked	● Green — double-blink every two seconds	White flash on each NTP request served
GPS holdover	● Amber — triple-blink every two seconds	White flash on each NTP request served
No sync	Off	Off
AP setup mode	Alternating white, 500 ms	Alternating white, opposite the STATUS LED
Firmware update	Solid white	Solid white
IP-address readout	Blinks the IP out octet by octet (Section 8)	Alternates green/red with STATUS to mark each digit

The ACTIVITY LED gives a brief flash on each NTP request answered, so a healthy, busy server visibly ticks. The Indicator LEDs setting (Section 6) dims or disables the normal GPS, NTP, holdover, and activity indications; the attention states — AP setup mode, firmware update, and IP-address readout — always show at full brightness regardless of that setting.

Under-light (bottom)

The two bottom LEDs are decorative under-lighting set to a single constant color by the Under-light Color picker. Because the color carries its own brightness, picking a darker shade dims the under-light and choosing black turns it off entirely. The under-light is independent of the Indicator LEDs brightness setting.

8. The SW1 Button

Action	Result
Double-click	Blink out the server's IP address on the LEDs, one octet at a time
Hold 5 seconds	Switch to AP (setup) mode temporarily, to reconfigure WiFi or settings
Hold 10 seconds	Check for and install a software update over WiFi (see Section 9)
Hold during power-on	Factory reset — wipe all settings and reboot into AP setup mode

The two timed holds share one button, so the duration matters. Release at about five seconds for AP mode; keep holding to about ten seconds for the update check. Watch the LEDs: both go solid white when the update begins. A hold during power-on (before the LEDs come up) is the factory reset.

Reading the IP blink

After a double-click, the IP address is signalled octet by octet. Each digit is shown as that many blinks (a zero is shown as a single long flash), with the two LEDs alternating to mark digit boundaries, and a longer pause between octets. For example, 192.168.4.75 blinks as 1-9-2, then 1-6-8, then 4, then 7-5. For most users the status page is an easier way to read the address; the blink readout is there for when no screen is handy.

9. Over the Air Updates

New firmware can be installed over WiFi without a cable. Hold SW1 for ten seconds while the server is running. (A shorter, five-second hold switches to AP mode instead — see Section 8, so keep holding past five seconds for an update.) Both status LEDs go solid white while the update runs.

The server fetches a small version file from the configured Firmware URL and compares it with the running version. If they match, it reports that it is already current and resumes normal operation. If a newer version is available, it downloads and installs it, then reboots into the new firmware. The whole process is automatic once triggered; do not remove power while both LEDs are solid white.

The Firmware URL is set on the System sub-page and points at the location that hosts the version file and firmware image. The Update Firmware button on that page does the same thing as the 10-second button hold, from the comfort of the web interface.

10. Specifications

Item	Specification
Controller	ESP32-S3 (240 MHz dual-core, WiFi)
Time source	DS3231 TCXO RTC (± 2 ppm), battery-backed
Reference (GPS mode)	u-blox GPS, NMEA + 1PPS — stratum 1
Reference (NTP mode)	Upstream NTP server — stratum 3
LAN accuracy (GPS)	Approximately ± 1 ms served; clients typically see ± 15 ms end to end over WiFi
Holdover drift	About 7 ms per hour on the DS3231 after GPS loss
Protocol	NTP v4 (RFC 5905) over UDP port 123
Network	2.4 GHz WiFi; DHCP or static IP; mDNS discovery
Power	5 V USB-C

11. Troubleshooting

Symptom	What to check
Source stays at — / NO SYNC	In GPS mode, give the unit a clear view of the sky and allow several minutes for a first fix. Confirm WiFi is connected so the NTP fallback can also work.
Never gets a GPS fix	Move the unit to a window or outdoors; cold starts under cover can take a long time or never complete. The status page Source will read NTP while GPS is unacquired.
Status shows HOLDOVER	GPS lost its fix but the server is still serving GPS-derived time from the RTC. If it persists, improve the unit's view of the sky; after the NTP Fallback window the server switches to NTP.
GPS-NTP shown in red	GPS and the upstream NTP server disagree by over a second. Suspect a GPS fault, poor reception, or a bad upstream server; try an alternate NTP server.
Clients can't reach the server	Confirm the server's IP on the status page and that clients point at it. A DHCP reservation or static IP keeps the address stable.
Can't find the server on the network	Try <code>http://ntpserver.local/</code> (or the configured name). If WiFi setup failed, the unit falls back to AP mode — rejoin the NTPServer network and reconfigure.
Both LEDs alternate fast and won't stop	The unit is not provisioned (authenticity record missing). It must be provisioned before the firmware will run.
Forgotten settings / start over	Hold SW1 while applying power to factory reset into AP mode.

Notices

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Power the server only from a standard 5 V DC USB-C source. The NTPServer contains no user-serviceable parts.

NTPServer — Designed by Mitchell Feig